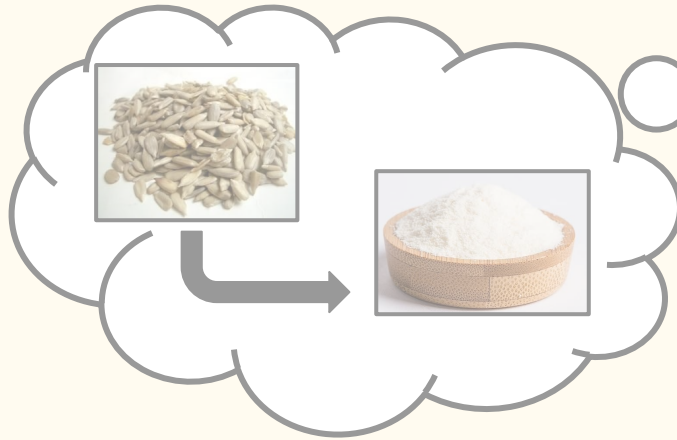


Parte 4

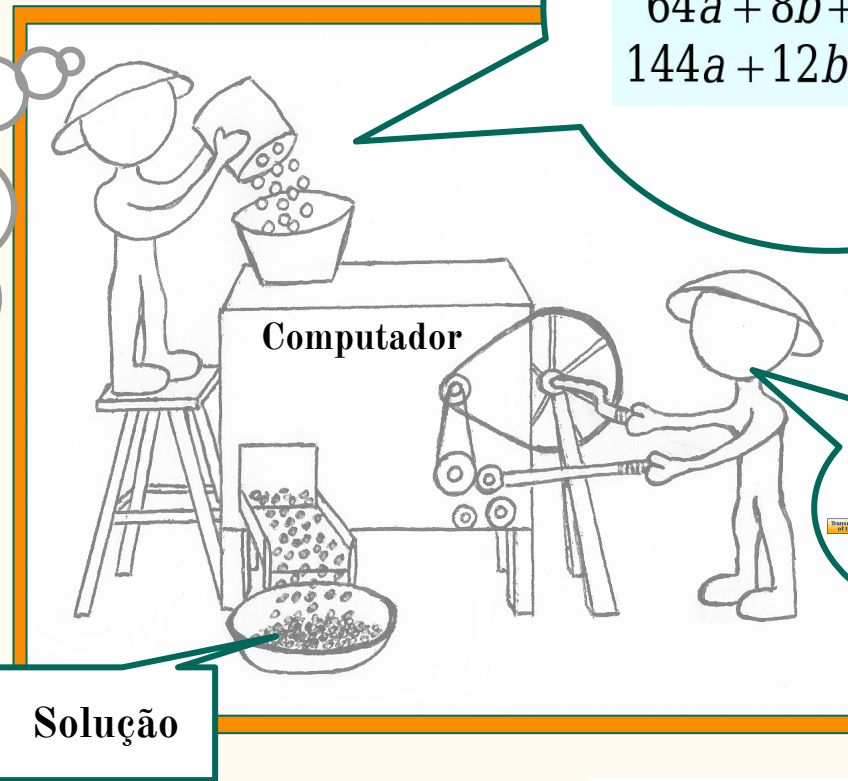
Resolução de Sistemas Lineares - 2

CI1164 - Introdução à Computação Científica
Profs. Armando Delgado e Guilherme Derenievicz
Departamento de Informática - UFPR

Visão geral da disciplina:



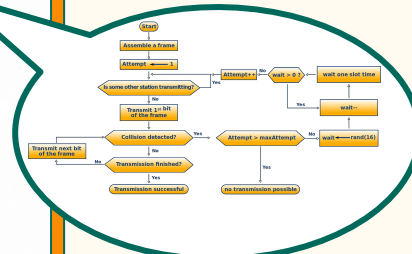
Descrição do Problema



Solução

$$\begin{aligned}25a + 5b + c &= 106 \\64a + 8b + c &= 177 \\144a + 12b + c &= 600\end{aligned}$$

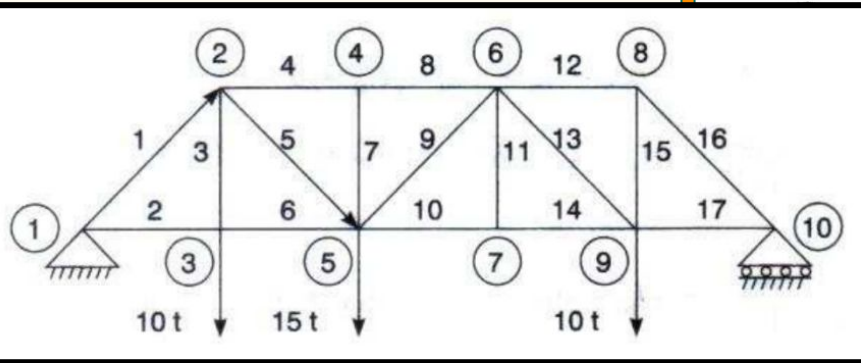
Sistemas Lineares



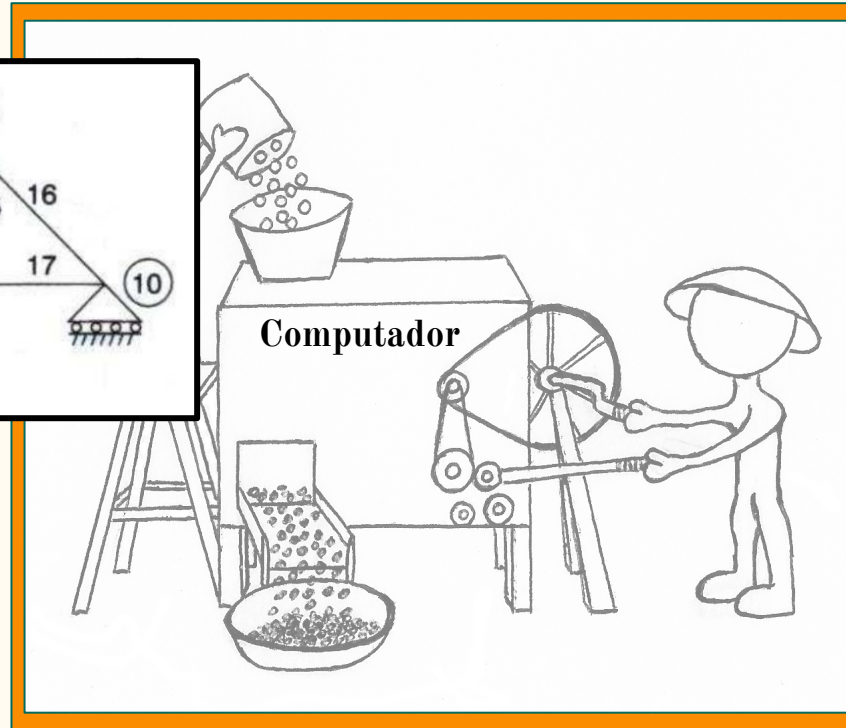
Método Numérico

Resolução de Sistemas Lineares

Fonte: Cálculo Numérico (Ruggiero & Lopes)

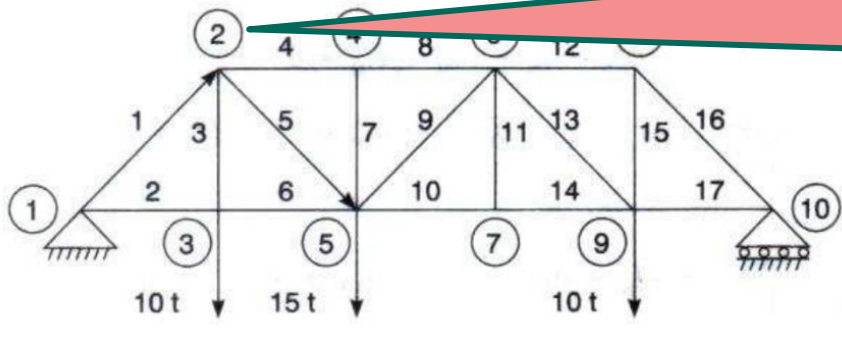


Descrição do Problema



Resolução de Sistemas Lineares

Fonte: Cálculo Numérico (Ruggiero & Lopes)



Descrição do Problema

$$\begin{cases} -\alpha f_1 + f_4 + \alpha f_5 = 0 \\ -\alpha f_1 - f_3 - \alpha f_5 = 0 \\ -f_2 + f_6 = 0 \\ f_3 - 10 = 0 \\ -f_4 + f_8 = 0 \\ -f_7 = 0 \\ -\alpha f_5 - f_6 + \alpha f_9 + f_{10} = 0 \\ \alpha f_5 + f_7 + \alpha f_9 - 15 = 0 \\ -f_8 - \alpha f_9 + f_{12} + \alpha f_{13} = 0 \\ -\alpha f_9 - f_{11} - \alpha f_{13} = 0 \\ -f_{10} + f_{14} = 0 \\ f_{11} = 0 \\ -f_{12} + \alpha f_{16} = 0 \\ -f_{15} - \alpha f_{16} = 0 \\ -\alpha f_{13} - f_{14} + f_{17} = 0 \\ \alpha f_{13} + f_{15} - f_{10} = 0 \\ \alpha f_{16} - f_{17} = 0 \end{cases}$$

$$\alpha f_5 = 0$$

$$\alpha f_5 = 0$$

Fatoração LU

Problema: calcular a inversa de uma matriz.

$$\begin{array}{c} \mathbf{A} \\ \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{array} \right] \end{array} \begin{array}{c} \mathbf{A}^{-1} \\ \left[\begin{array}{cccc} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ x_{41} & x_{42} & x_{43} & x_{44} \end{array} \right] \end{array} = \begin{array}{c} \mathbf{I} \\ \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right] \end{array}$$

Fatoração LU

$$a_{11}x_{11} + a_{12}x_{21} + a_{13}x_{31} + a_{14}x_{41} = 1$$

Problema: calcular a inversa de uma matriz.

$$\begin{matrix} & \mathbf{A} & & \mathbf{A}^{-1} & & \mathbf{I} \end{matrix}$$
$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ x_{41} & x_{42} & x_{43} & x_{44} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Fatoração LU

Problema: calcular a inversa de uma matriz.

$$\begin{aligned}a_{11}x_{11} + a_{12}x_{21} + a_{13}x_{31} + a_{14}x_{41} &= 1 \\a_{21}x_{11} + a_{22}x_{21} + a_{23}x_{31} + a_{24}x_{41} &= 0\end{aligned}$$

$$\begin{matrix} & \mathbf{A} & & \mathbf{A}^{-1} & & \mathbf{I} \end{matrix}$$
$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ x_{41} & x_{42} & x_{43} & x_{44} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Fatoração LU

Problema: calcular a inversa de uma matriz.

$$a_{11}x_{11} + a_{12}x_{21} + a_{13}x_{31} + a_{14}x_{41} = 1$$

$$a_{21}x_{11} + a_{22}x_{21} + a_{23}x_{31} + a_{24}x_{41} = 0$$

$$a_{31}x_{11} + a_{32}x_{21} + a_{33}x_{31} + a_{34}x_{41} = 0$$

$$\begin{matrix} & \mathbf{A} & & \mathbf{A}^{-1} & & \mathbf{I} \end{matrix}$$
$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ x_{41} & x_{42} & x_{43} & x_{44} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Fatoração LU

Problema: calcular a inversa de uma matriz.

$$\begin{cases} a_{11}x_{11} + a_{12}x_{21} + a_{13}x_{31} + a_{14}x_{41} = 1 \\ a_{21}x_{11} + a_{22}x_{21} + a_{23}x_{31} + a_{24}x_{41} = 0 \\ a_{31}x_{11} + a_{32}x_{21} + a_{33}x_{31} + a_{34}x_{41} = 0 \\ a_{41}x_{11} + a_{42}x_{21} + a_{43}x_{31} + a_{44}x_{41} = 0 \end{cases}$$

$$\begin{matrix} & \mathbf{A} & & \mathbf{A}^{-1} & & \mathbf{I} \end{matrix}$$
$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ x_{41} & x_{42} & x_{43} & x_{44} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Fatoração LU

Problema: calcular a inversa de uma matriz.

$$\begin{cases} a_{11}x_{11} + a_{12}x_{21} + a_{13}x_{31} + a_{14}x_{41} = 1 \\ a_{21}x_{11} + a_{22}x_{21} + a_{23}x_{31} + a_{24}x_{41} = 0 \\ a_{31}x_{11} + a_{32}x_{21} + a_{33}x_{31} + a_{34}x_{41} = 0 \\ a_{41}x_{11} + a_{42}x_{21} + a_{43}x_{31} + a_{44}x_{41} = 0 \end{cases}$$

$$\begin{matrix} & \mathbf{A} & & \mathbf{A}^{-1} & & \mathbf{I} \end{matrix}$$
$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ x_{41} & x_{42} & x_{43} & x_{44} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{A}\mathbf{x}_1 = \mathbf{i}_1$$

Fatoração LU

Problema: calcular a inversa de uma matriz.

$$\begin{cases} a_{11}x_{11} + a_{12}x_{21} + a_{13}x_{31} + a_{14}x_{41} = 1 \\ a_{21}x_{11} + a_{22}x_{21} + a_{23}x_{31} + a_{24}x_{41} = 0 \\ a_{31}x_{11} + a_{32}x_{21} + a_{33}x_{31} + a_{34}x_{41} = 0 \\ a_{41}x_{11} + a_{42}x_{21} + a_{43}x_{31} + a_{44}x_{41} = 0 \end{cases}$$

$$\begin{matrix} & \mathbf{A} & & \mathbf{A}^{-1} & & \mathbf{I} & & \end{matrix}$$
$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ x_{41} & x_{42} & x_{43} & x_{44} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$Ax_1 = i_1$

$Ax_2 = i_2$

$Ax_3 = i_3$

$Ax_4 = i_4$

Fatoração LU

Decompor $\mathbf{A}$ em duas matrizes triangulares $\mathbf{A} = \mathbf{L}\mathbf{U}$.

Fatoração LU

Decompor $\mathbf{A}$ em duas matrizes triangulares $\mathbf{A} = \mathbf{L}\mathbf{U}$.

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

Fatoração LU

Decompor $\mathbf{A}$ em duas matrizes triangulares $\mathbf{A} = \mathbf{LU}$.

$$\mathbf{Ax} = \mathbf{b}$$

$$\mathbf{LUx} = \mathbf{b}$$

Fatoração LU

Decompor $\mathbf{A}$ em duas matrizes triangulares $\mathbf{A} = \mathbf{LU}$.

$$\mathbf{Ax} = \mathbf{b}$$

$$\underbrace{\mathbf{LU}}_{\mathbf{y}} \mathbf{x} = \mathbf{b}$$

Fatoração LU

Decompor $\mathbf{A}$ em duas matrizes triangulares $\mathbf{A} = \mathbf{LU}$.

$$\mathbf{Ax} = \mathbf{b}$$

$$\underbrace{\mathbf{LU}}_{\mathbf{y}} \mathbf{x} = \mathbf{b} \quad \longrightarrow \quad \mathbf{Ly} = \mathbf{b}$$
$$\mathbf{Ux} = \mathbf{y}$$

Fatoração LU

Decompor $\mathbf{A}$ em duas matrizes triangulares $\mathbf{A} = \mathbf{LU}$.

$$\mathbf{Ax} = \mathbf{b}$$

$$\underbrace{\mathbf{LU}}_{\mathbf{y}} \mathbf{x} = \mathbf{b}$$



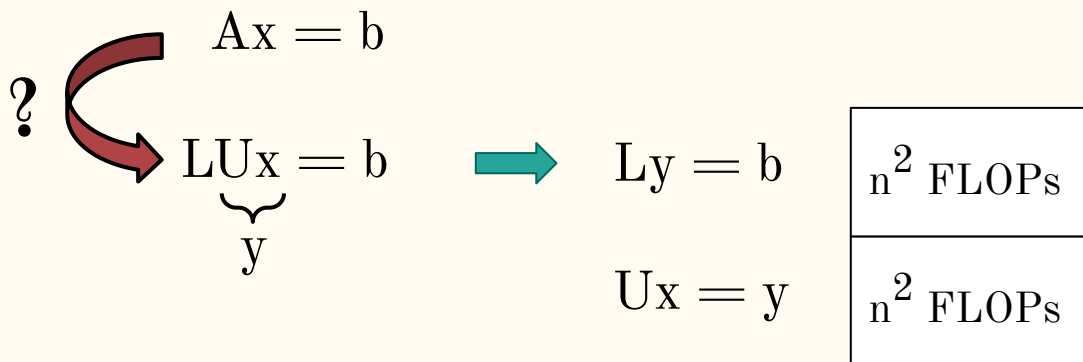
$$\mathbf{Ly} = \mathbf{b}$$

$$\mathbf{Ux} = \mathbf{y}$$

n^2 FLOPs
n^2 FLOPs

Fatoração LU

Decompor $\mathbf{A}$ em duas matrizes triangulares $\mathbf{A} = \mathbf{LU}$.



Fatoração LU

Decompor $\mathbf{A}$ em duas matrizes triangulares $\mathbf{A} = \mathbf{LU}$.

$$\mathbf{A}\mathbf{x}_1 = \mathbf{i}_1$$

$$\mathbf{A}\mathbf{x}_2 = \mathbf{i}_2$$

$$\mathbf{A}\mathbf{x}_3 = \mathbf{i}_3$$

$\vdots$

$$\mathbf{A}\mathbf{x}_k = \mathbf{i}_k$$

?


$$\mathbf{A}\mathbf{x} = \mathbf{b}$$
$$\underbrace{\mathbf{L}\mathbf{U}}_{\mathbf{y}}\mathbf{x} = \mathbf{b}$$



$$\mathbf{L}\mathbf{y} = \mathbf{b}$$

$$\mathbf{U}\mathbf{x} = \mathbf{y}$$

n^2 FLOPs

n^2 FLOPs

$\text{custo}(\text{decomposição}) + 2kn^2 \text{ FLOPs}$

Fatoração LU

Decompor $\mathbf{A}$ em duas matrizes triangulares $\mathbf{A} = \mathbf{L}\mathbf{U}$.

$\mathbf{A}$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

Fatoração LU

Decompor $\mathbf{A}$ em duas matrizes triangulares $\mathbf{A} = \mathbf{L}\mathbf{U}$.

 m_{21} m_{31} m_{41} $\mathbf{A}$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{matrix} \\ m_{21} \\ m_{31} \\ m_{41} \end{matrix}$$

 $\mathbf{A}_1$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & b_{22} & b_{23} & b_{24} \\ 0 & b_{32} & b_{33} & b_{34} \\ 0 & b_{42} & b_{43} & b_{44} \end{bmatrix}$$

Fatoração LU

Decompor $\mathbf{A}$ em duas matrizes triangulares $\mathbf{A} = \mathbf{LU}$.

$$\begin{matrix} m_{21} \\ m_{31} & m_{32} \\ m_{41} & m_{42} \end{matrix}$$

$$\begin{matrix} \mathbf{A}_1 \\ \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & b_{22} & b_{23} & b_{24} \\ 0 & b_{32} & b_{33} & b_{34} \\ 0 & b_{42} & b_{43} & b_{44} \end{array} \right] \begin{matrix} m_{32} \\ m_{42} \end{matrix} \end{matrix} \quad \rightarrow \quad \begin{matrix} \mathbf{A}_2 \\ \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & b_{22} & b_{23} & b_{24} \\ 0 & 0 & c_{33} & c_{34} \\ 0 & 0 & c_{43} & c_{44} \end{array} \right] \end{matrix}$$

Fatoração LU

Decompor $\mathbf{A}$ em duas matrizes triangulares $\mathbf{A} = \mathbf{LU}$.

$$\begin{array}{ccc} m_{21} & & \\ m_{31} & m_{32} & \\ m_{41} & m_{42} & m_{43} \end{array}$$

$$\mathbf{A}_2 = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & b_{22} & b_{23} & b_{24} \\ 0 & 0 & c_{33} & c_{34} \\ 0 & 0 & c_{43} & c_{44} \end{bmatrix} \quad m_{43}$$



$$\mathbf{A}_3 = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & b_{22} & b_{23} & b_{24} \\ 0 & 0 & c_{33} & c_{34} \\ 0 & 0 & 0 & d_{44} \end{bmatrix}$$

Fatoração LU

Decompor $\mathbf{A}$ em duas matrizes triangulares $\mathbf{A} = \mathbf{L}\mathbf{U}$.

$$\begin{array}{ccc} m_{21} & & \\ m_{31} & m_{32} & \\ m_{41} & m_{42} & m_{43} \end{array}$$

$$\begin{array}{c} \mathbf{A}_3 \\ \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ m_{21} & 1 & 0 & 0 \\ m_{31} & m_{32} & 1 & 0 \\ m_{41} & m_{42} & m_{43} & 1 \end{array} \right] \end{array} \begin{array}{c} \mathbf{A} \\ \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & b_{22} & b_{23} & b_{24} \\ 0 & 0 & c_{33} & c_{34} \\ 0 & 0 & 0 & d_{44} \end{array} \right] \end{array} = \begin{array}{c} \mathbf{A} \\ \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{array} \right] \end{array}$$

Fatoração LU

Decompor $\mathbf{A}$ em duas matrizes triangulares $\mathbf{A} = \mathbf{LU}$.

$$\begin{matrix} & & \mathbf{A}_3 & & \mathbf{A} \\ & & & & \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ m_{21} & 1 & 0 & 0 \\ m_{31} & m_{32} & 1 & 0 \\ m_{41} & m_{42} & m_{43} & 1 \end{bmatrix} & \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & b_{22} & b_{23} & b_{24} \\ 0 & 0 & c_{33} & c_{34} \\ 0 & 0 & 0 & d_{44} \end{bmatrix} & = & \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}
 \end{matrix}$$

$$\begin{aligned}
 m_{21}a_{12} + b_{22} &= (a_{21}/a_{11})a_{12} + (a_{22} - a_{12}m_{21}) \\
 &= (a_{12}a_{21})/a_{11} + a_{22} - (a_{12}a_{21})/a_{11}
 \end{aligned}$$

Fatoração LU

Decompor $\mathbf{A}$ em duas matrizes triangulares $\mathbf{A} = \mathbf{LU}$.

$$\begin{array}{c} \mathbf{L} \\ \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ m_{21} & 1 & 0 & 0 \\ m_{31} & m_{32} & 1 & 0 \\ m_{41} & m_{42} & m_{43} & 1 \end{array} \right] \end{array} \begin{array}{c} \mathbf{U} \\ \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & b_{22} & b_{23} & b_{24} \\ 0 & 0 & c_{33} & c_{34} \\ 0 & 0 & 0 & d_{44} \end{array} \right] \end{array} = \begin{array}{c} \mathbf{A} \\ \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{array} \right] \end{array}$$

Fatoração LU

Exemplo: encontrar a inversa da matriz **A**.

$$\mathbf{A} = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

Fatoração LU

Exemplo: encontrar a inversa da matriz **A**.

$$\begin{array}{c} \mathbf{A} \\ \left[\begin{array}{ccc} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{array} \right] \end{array} \begin{array}{c} \mathbf{X} \\ \left[\begin{array}{ccc} \mathbf{x}_{11} & \mathbf{x}_{12} & \mathbf{x}_{13} \\ \mathbf{x}_{21} & \mathbf{x}_{22} & \mathbf{x}_{23} \\ \mathbf{x}_{31} & \mathbf{x}_{32} & \mathbf{x}_{33} \end{array} \right] \end{array} = \begin{array}{c} \mathbf{B} \\ \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \end{array}$$

Fatoração LU

Exemplo: encontrar a inversa da matriz **A**.

$$\begin{matrix} & \mathbf{A} \\ \left[\begin{array}{ccc} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{array} \right] & \begin{array}{l} m_{21} = 2.56 \\ m_{31} = 5.76 \end{array} \end{matrix}$$

Fatoração LU

Exemplo: encontrar a inversa da matriz **A**.

$$\begin{array}{c} \mathbf{A} \\ \left[\begin{array}{ccc} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{array} \right] \end{array} \begin{array}{l} m_{21} = 2.56 \\ m_{31} = 5.76 \end{array} \rightarrow \left[\begin{array}{ccc} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & -16.8 & -4.76 \end{array} \right]$$

Fatoração LU

Exemplo: encontrar a inversa da matriz **A**.

$$\begin{array}{c} \mathbf{A} \\ \left[\begin{array}{ccc} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{array} \right] \begin{array}{l} m_{21} = 2.56 \\ m_{31} = 5.76 \end{array} \end{array} \rightarrow \begin{array}{c} \left[\begin{array}{ccc} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & -16.8 & -4.76 \end{array} \right] \begin{array}{l} \\ m_{32} = 3.5 \end{array} \end{array}$$

Fatoração LU

Exemplo: encontrar a inversa da matriz **A**.

$$\begin{array}{c} \mathbf{A} \\ \left[\begin{array}{ccc} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{array} \right] \begin{array}{l} m_{21} = 2.56 \\ m_{31} = 5.76 \end{array} \end{array} \quad \longrightarrow \quad \begin{array}{c} \left[\begin{array}{ccc} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & -16.8 & -4.76 \end{array} \right] m_{32} = 3.5 \end{array}$$

$$\longrightarrow \left[\begin{array}{ccc} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{array} \right]$$

Fatoração LU

Exemplo: encontrar a inversa da matriz **A**.

$$\begin{array}{c} \mathbf{A} \\ \left[\begin{array}{ccc} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{array} \right] \begin{array}{l} m_{21} = 2.56 \\ m_{31} = 5.76 \end{array} \end{array} \quad \longrightarrow \quad \begin{array}{c} \left[\begin{array}{ccc} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & -16.8 & -4.76 \end{array} \right] m_{32} = 3.5 \end{array}$$

$$\longrightarrow \begin{array}{c} \left[\begin{array}{ccc} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{array} \right] \\ \mathbf{U} \end{array}$$

Fatoração LU

Exemplo: encontrar a inversa da matriz **A**.

$$\begin{array}{c} \mathbf{A} \\ \left[\begin{array}{ccc} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{array} \right] \begin{array}{l} m_{21} = 2.56 \\ m_{31} = 5.76 \end{array} \end{array} \longrightarrow \begin{array}{c} \left[\begin{array}{ccc} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & -16.8 & -4.76 \end{array} \right] m_{32} = 3.5 \end{array}$$

$$\longrightarrow \begin{array}{c} \left[\begin{array}{ccc} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{array} \right] \\ \mathbf{U} \end{array} \quad \begin{array}{c} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{array} \right] \\ \mathbf{L} \end{array}$$

Fatoração LU

Exemplo: encontrar a inversa da matriz **A**.

$$Ax = b$$

$$L\underset{y}{U}x = b \quad \longrightarrow \quad Ly = b$$

$$Ux = y$$

$$\begin{matrix} & \mathbf{A} \\ \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} & \begin{matrix} m_{21} = 2.56 \\ m_{31} = 5.76 \end{matrix} \end{matrix} \quad \longrightarrow \quad \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & -16.8 & -4.76 \end{bmatrix} \begin{matrix} \\ \\ m_{32} = 3.5 \end{matrix}$$

$$\begin{matrix} \longrightarrow & \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} & \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \\ & \mathbf{U} & \mathbf{L} \end{matrix}$$

Fatoração LU

Exemplo: encontrar a inversa da matriz **A**.

$$Ax = b$$

$$L \underbrace{Ux}_y = b \quad \rightarrow \quad Ly = b$$

$$Ux = y$$

L

$$\begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \rightarrow \quad \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -2.56 \\ 3.2 \end{bmatrix}$$

Aplica o mesmo
escalamento
no vetor b

U

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -2.56 \\ 3.2 \end{bmatrix} \quad \rightarrow \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0.04762 \\ -0.9524 \\ 4.571 \end{bmatrix}$$

Fatoração LU

Exemplo: encontrar a inversa da matriz **A**.

$$Ax = b$$

$$L \underbrace{U}_y x = b \quad \rightarrow \quad Ly = b$$

$$Ux = y$$

L

$$\begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$



$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -3.5 \end{bmatrix}$$

Aplica o mesmo
escalonamento
no vetor b

U

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -3.5 \end{bmatrix}$$



$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -0.08333 \\ 1.417 \\ -5.000 \end{bmatrix}$$

Fatoração LU

Exemplo: encontrar a inversa da matriz **A**.

$$Ax = b$$

$$L \underbrace{Ux}_y = b \quad \rightarrow \quad Ly = b$$

$$Ux = y$$

L

$$\begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$



$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Aplica o mesmo
escalonamento
no vetor b

U

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$



$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0.03571 \\ -0.4643 \\ 1.429 \end{bmatrix}$$

Fatoração LU

Exemplo: encontrar a inversa da matriz A .

$$A^{-1}$$

$$\begin{bmatrix} 0.04762 & -0.08333 & 0.03571 \\ -0.9524 & 1.417 & -0.4643 \\ 4.571 & -5.000 & 1.429 \end{bmatrix}$$

$$Ax = b$$

$$L \underbrace{Ux}_y = b \quad \Rightarrow \quad Ly = b$$

$$Ux = y$$

Fatoração LU

Exemplo: encontrar a inversa da matriz $\mathbf{A}$.

Para pivoteamento parcial, efetuar as mesmas trocas de linhas no vetor $\mathbf{b}$.

$$\mathbf{Ax} = \mathbf{b}$$

$$\mathbf{L}\underbrace{\mathbf{Ux}}_{\mathbf{y}} = \mathbf{b} \quad \longrightarrow \quad \mathbf{Ly} = \mathbf{b}$$

$$\mathbf{Ux} = \mathbf{y}$$

Fatoração LU

Exemplo: encontrar a inversa da matriz **A**.

Para pivoteamento parcial, efetuar as mesmas trocas de linhas no vetor **b**.

$$Ax = b$$

$$L\underset{y}{U}x = b \quad \longrightarrow \quad Ly = b$$

$$Ux = y$$

Exemplo:

$$\begin{matrix} & A & & A' \\ \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} & \longrightarrow & \begin{bmatrix} 144 & 12 & 1 \\ 64 & 8 & 1 \\ 25 & 5 & 1 \end{bmatrix} & \longrightarrow & A' = LU \end{matrix}$$

Fatoração LU

$$Ax = b$$

Exemplo: encontrar a inversa da matriz A

Para pivoteamento parcial, efetuar as m de linhas no vetor b .

$$LU \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Exemplo:

$$A = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$



$$A' = \begin{bmatrix} 144 & 12 & 1 \\ 64 & 8 & 1 \\ 25 & 5 & 1 \end{bmatrix}$$



$$A' = LU$$

Fatoração LU

Exemplo: encontrar a inversa da matriz A

Para pivoteamento parcial, efetuar as m de linhas no vetor $\mathbf{b}$.

Exemplo:

$$A = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$



$$A' = \begin{bmatrix} 144 & 12 & 1 \\ 64 & 8 & 1 \\ 25 & 5 & 1 \end{bmatrix}$$



$$A' = LU$$

$$Ax = b$$

LU

$$\begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Atenção: além do vetor $\mathbf{b}$, as mesmas trocas de linhas devem ser feitas a cada iteração na matriz L .

Fatoração LU

Exemplo:

$$\begin{matrix} & \mathbf{A} & & \mathbf{x} & & \mathbf{b} \\ \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} & \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} & = & \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \end{matrix}$$

Fatoração LU

Exemplo:

$$A = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

L

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$$

b

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Fatoração LU

Exemplo:

$$A = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

Escolhendo o primeiro pivô:
trocar as linhas 1 e 3.

L

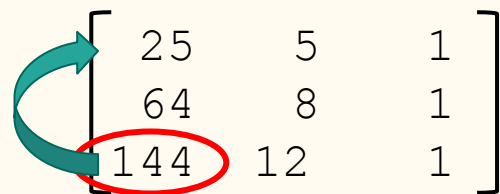
$$\begin{bmatrix} \\ \\ \end{bmatrix}$$

b

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Fatoração LU

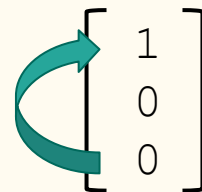
Exemplo:

$$A = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$


L

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$$

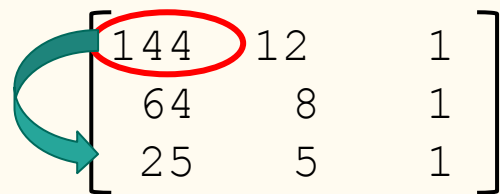
b

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$


A mesma troca deve ser feita no vetor **b**.

Fatoração LU

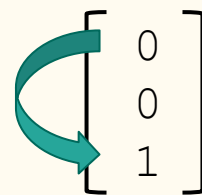
Exemplo:

$$A = \begin{bmatrix} 144 & 12 & 1 \\ 64 & 8 & 1 \\ 25 & 5 & 1 \end{bmatrix}$$


L

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$$

b

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$


A mesma troca deve ser feita no vetor **b**.

Fatoração LU

Exemplo:

$$\begin{array}{c} A \\ \left[\begin{array}{cc|c} 144 & 12 & 1 \\ 64 & 8 & 1 \\ 25 & 5 & 1 \end{array} \right] \end{array} \quad \begin{array}{l} m_{21} = 0.4444 \\ m_{31} = 0.1736 \end{array}$$

Definindo multiplicadores
da primeira iteração.

L

$$\left[\begin{array}{c} \\ \\ \\ \end{array} \right]$$

b

$$\left[\begin{array}{c} 0 \\ 0 \\ 1 \end{array} \right]$$

Fatoração LU

Exemplo:

$$\begin{array}{c} A \\ \left[\begin{array}{cc|c} 144 & 12 & 1 \\ 64 & 8 & 1 \\ 25 & 5 & 1 \end{array} \right] \end{array} \quad \begin{array}{l} m_{21} = 0.4444 \\ m_{31} = 0.1736 \end{array}$$

Armazenando na matriz L.

$$\begin{array}{c} L \\ \left[\begin{array}{c} 0.4444 \\ 0.1736 \end{array} \right] \end{array}$$

$$\begin{array}{c} b \\ \left[\begin{array}{c} 0 \\ 0 \\ 1 \end{array} \right] \end{array}$$

Fatoração LU

Exemplo:

$$\begin{array}{ccc} & \mathbf{A} & \\ \left[\begin{array}{cc|c} 144 & 12 & 1 \\ 64 & 8 & 1 \\ 25 & 5 & 1 \end{array} \right] & \begin{array}{l} m_{21} = 0.4444 \\ m_{31} = 0.1736 \end{array} & \end{array}$$



Aplicando operações sobre as equações.

$$\left[\begin{array}{cc|c} 144 & 12 & 1 \\ 0 & 2.6672 & 0.5556 \\ 0 & 2.9168 & 0.8264 \end{array} \right]$$

$\mathbf{L}$

$$\left[\begin{array}{c} 0.4444 \\ 0.1736 \end{array} \right]$$

$\mathbf{b}$

$$\left[\begin{array}{c} 0 \\ 0 \\ 1 \end{array} \right]$$

Fatoração LU

Exemplo:

A

$$\begin{bmatrix} 144 & 12 & 1 \\ 0 & 2.6672 & 0.5556 \\ 0 & 2.9168 & 0.8264 \end{bmatrix}$$

Na segunda iteração é necessário trocar as linhas 2 e 3.

L

$$\begin{bmatrix} 0.4444 \\ 0.1736 \end{bmatrix}$$

b

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Fatoração LU

Exemplo:

A


$$\begin{bmatrix} 144 & 12 & 1 \\ 0 & 2.6672 & 0.5556 \\ 0 & 2.9168 & 0.8264 \end{bmatrix}$$

L


$$\begin{bmatrix} 0.4444 \\ 0.1736 \end{bmatrix}$$

A mesma troca deve ser feita no vetor **b** e nos elementos já preenchidos da matriz **L**.

b


$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Fatoração LU

Exemplo:

A


$$\begin{bmatrix} 144 & 12 & 1 \\ 0 & 2.9168 & 0.8264 \\ 0 & 2.6672 & 0.5556 \end{bmatrix}$$

A mesma troca deve ser feita no vetor **b** e nos elementos já preenchidos da matriz **L**.

L


$$\begin{bmatrix} 0.1736 \\ 0.4444 \end{bmatrix}$$

b


$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Fatoração LU

Exemplo:

A

$$\begin{bmatrix} 144 & 12 & 1 \\ 0 & 2.9168 & 0.8264 \\ 0 & 2.6672 & 0.5556 \end{bmatrix} m_{32} = 0.9144$$

Definindo multiplicadores
da segunda iteração.

L

$$\begin{bmatrix} 0.1736 \\ 0.4444 \end{bmatrix}$$

b

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Fatoração LU

Exemplo:

A

$$\begin{bmatrix} 144 & 12 & 1 \\ 0 & 2.9168 & 0.8264 \\ 0 & 2.6672 & 0.5556 \end{bmatrix} m_{32} = 0.9144$$

Armazenando na matriz L.

L

$$\begin{bmatrix} 0.1736 & & \\ 0.4444 & 0.9144 & \end{bmatrix}$$

b

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Fatoração LU

Exemplo:

$$A = \begin{bmatrix} 144 & 12 & 1 \\ 0 & 2.9168 & 0.8264 \\ 0 & 0 & -0.2001 \end{bmatrix}$$



Aplicando operações sobre as equações.

$$\begin{bmatrix} 144 & 12 & 1 \\ 0 & 2.9168 & 0.8264 \\ 0 & 2.6672 & 0.5556 \end{bmatrix} m_{32} = 0.9144$$

L

$$\begin{bmatrix} 0.1736 & & \\ 0.4444 & 0.9144 & \end{bmatrix}$$

b

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Fatoração LU

Exemplo:

A

$$\begin{bmatrix} 144 & 12 & 1 \\ 0 & 2.9168 & 0.8264 \\ 0 & 0 & -0.2001 \end{bmatrix}$$

U



Completando matriz L.

L

$$\begin{bmatrix} 1 & 0 & 0 \\ 0.1736 & 1 & 0 \\ 0.4444 & 0.9144 & 1 \end{bmatrix}$$

b

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Na prática: efetuar as trocas no(s) vetor(es) **b** posteriormente. No momento da fatoração, apenas armazenar as trocas em uma estrutura de dados.

Métodos de Resolução de Sistemas Lineares

- **Métodos Exatos ou Diretos:** permitiram a solução exata com um número finito de operações, se não fosse por erros numéricos.
- **Métodos Iterativos:** permitem uma solução aproximada através de um processo infinito convergente.

O Método de Gauss-Jacobi

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots + a_{nn}x_n = b_n \end{cases}$$

O Método de Gauss-Jacobi

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots + a_{nn}x_n = b_n \end{cases}$$

O Método de Gauss-Jacobi

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots + a_{nn}x_n = b_n \end{cases}$$

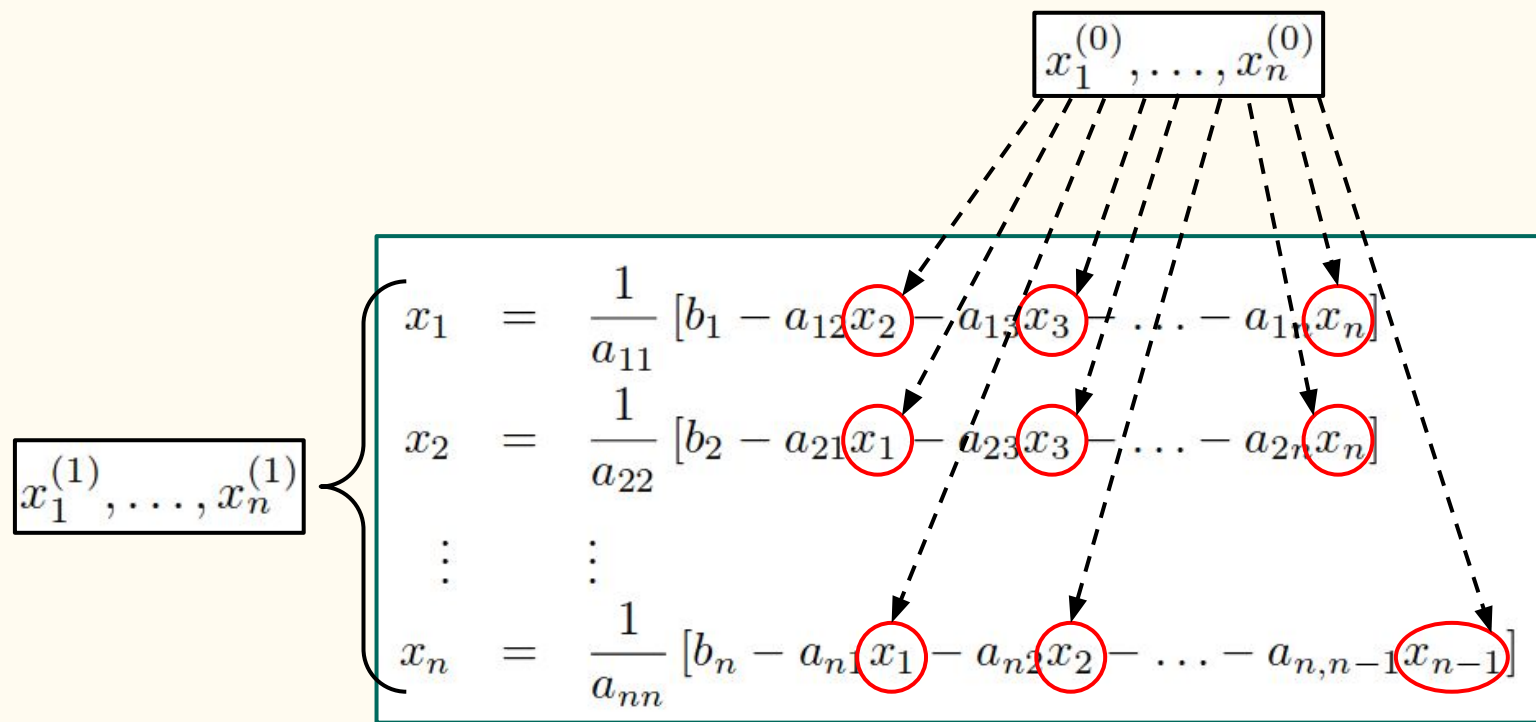
$$\begin{aligned} x_1 &= \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\ x_2 &= \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\ &\vdots \\ x_n &= \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}] \end{aligned}$$

O Método de Gauss-Jacobi

$$x_1^{(0)}, \dots, x_n^{(0)}$$

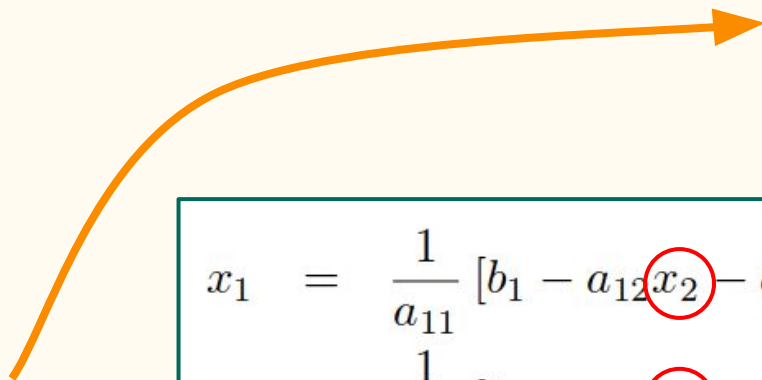
$$\begin{aligned} x_1 &= \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\ x_2 &= \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\ &\vdots \\ x_n &= \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}] \end{aligned}$$

O Método de Gauss-Jacobi



O Método de Gauss-Jacobi

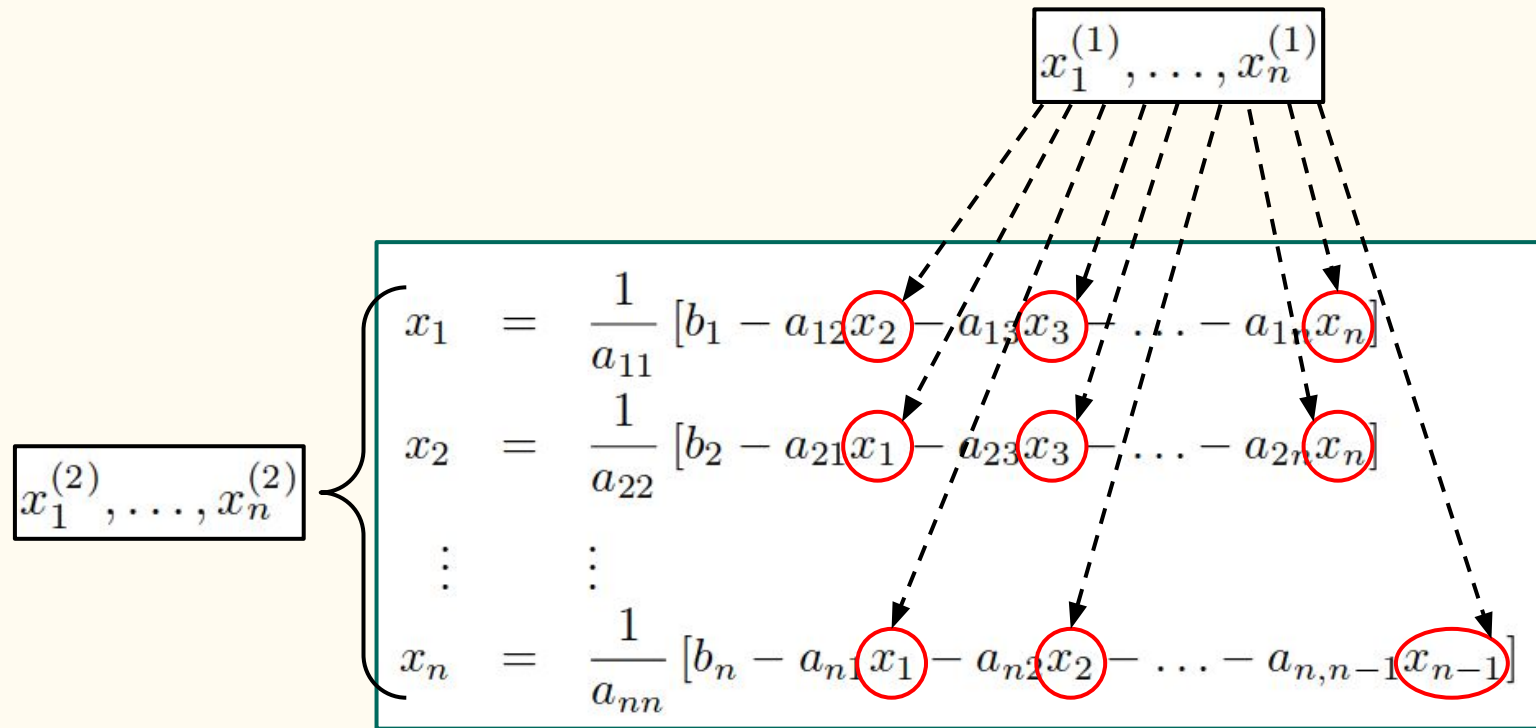
$$x_1^{(1)}, \dots, x_n^{(1)}$$



$$x_1^{(1)}, \dots, x_n^{(1)}$$

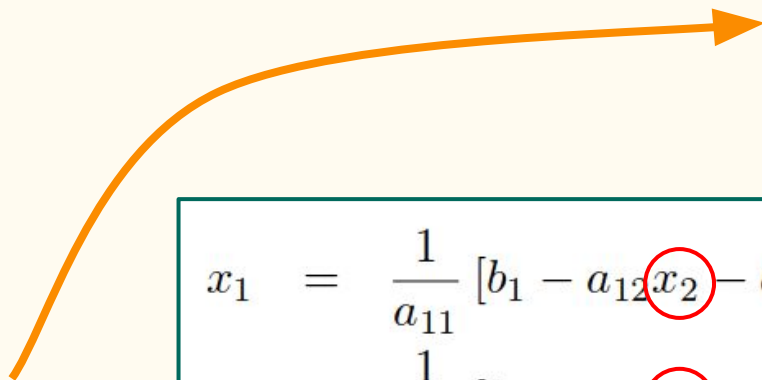
$$\begin{aligned} x_1 &= \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\ x_2 &= \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\ &\vdots \\ x_n &= \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}] \end{aligned}$$

O Método de Gauss-Jacobi



O Método de Gauss-Jacobi

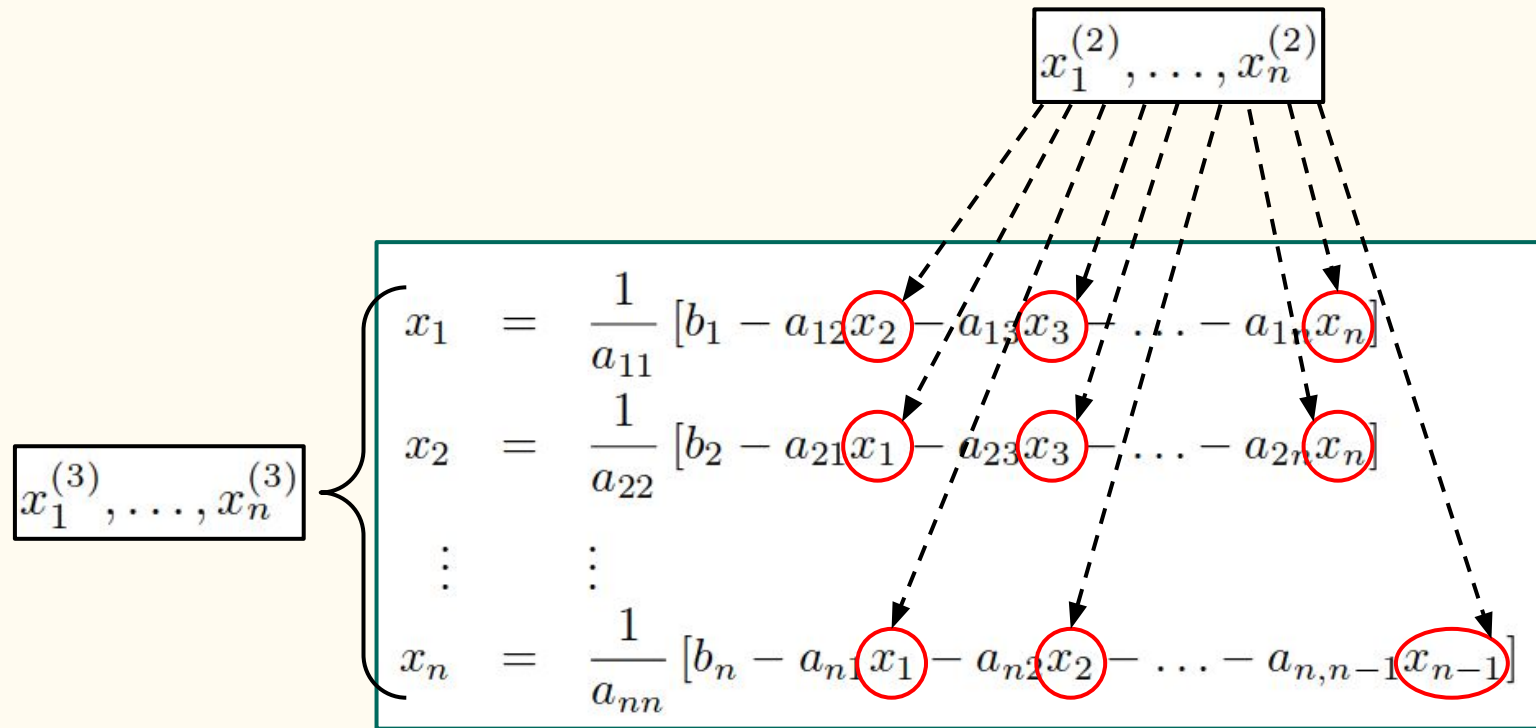
$$x_1^{(2)}, \dots, x_n^{(2)}$$



$$x_1^{(2)}, \dots, x_n^{(2)}$$

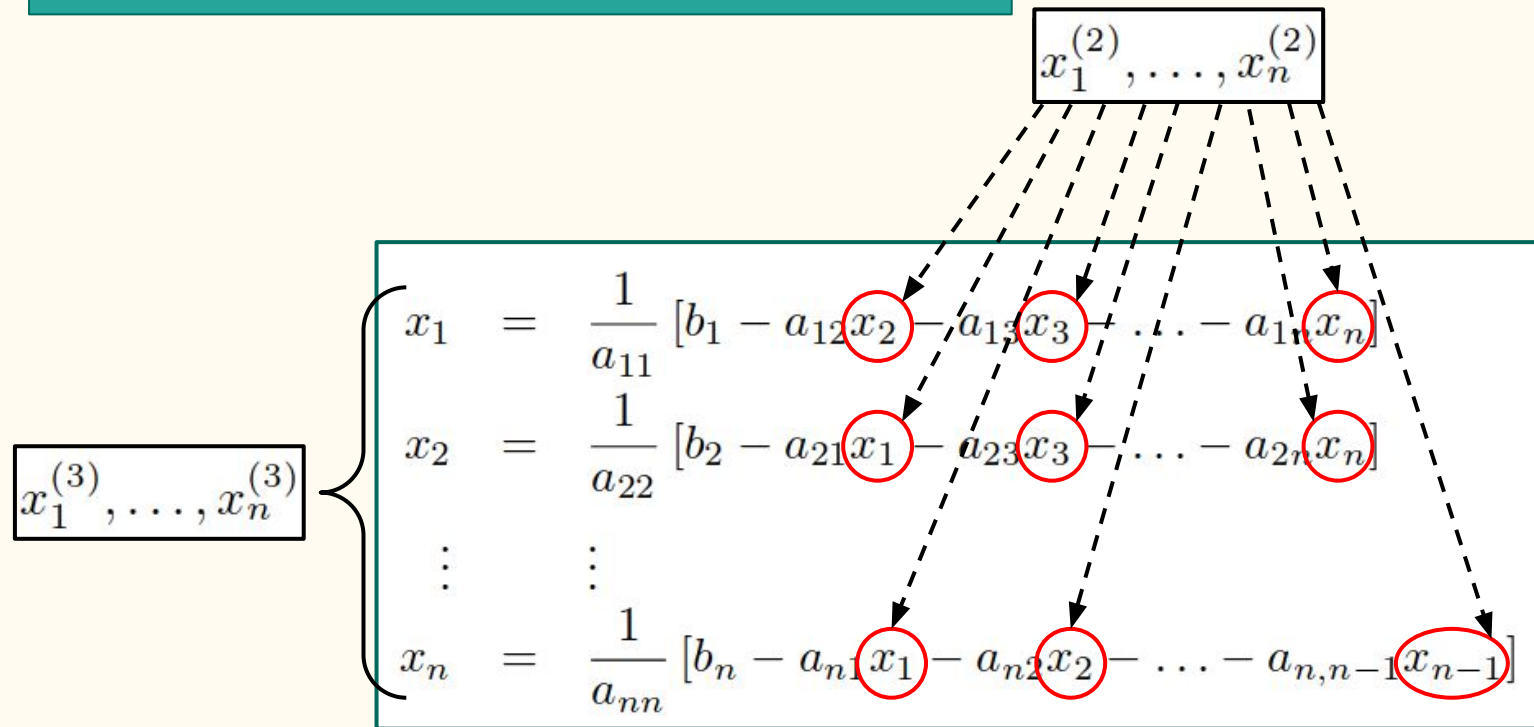
$$\begin{aligned} x_1 &= \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\ x_2 &= \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\ &\vdots \\ x_n &= \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}] \end{aligned}$$

O Método de Gauss-Jacobi



O Método de Gauss-Jacobi

$$\text{Max } \left\{ \left| x_i^{(k+1)} - x_i^{(k)} \right| \right\} \leq \varepsilon \quad i = 1, 2, 3, \dots, n$$



O Método de Gauss-Jacobi

$$\text{Max} \left\{ \left| \frac{x_i^{(k+1)} - x_i^{(k)}}{x_i^{(k+1)}} \right| \right\} \leq \varepsilon \quad i = 1, 2, 3, \dots, n$$

$$x_1^{(2)}, \dots, x_n^{(2)}$$

$$x_1^{(3)}, \dots, x_n^{(3)}$$

$$\begin{cases} x_1 = \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\ x_2 = \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\ \vdots \\ x_n = \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}] \end{cases}$$

O Método de Gauss-Jacobi

$$\text{Max } \left\{ \left| r_i^{(k+1)} \right| \right\} \leq \varepsilon \quad i = 1, 2, 3, \dots, n$$

$$r_i^{(k+1)} = b_i - \sum_{j=1}^n a_{ij} x_j^{(k+1)}$$

$$x_1^{(2)}, \dots, x_n^{(2)}$$

$$x_1^{(3)}, \dots, x_n^{(3)}$$

$$\begin{cases} x_1 = \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\ x_2 = \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\ \vdots \\ x_n = \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}] \end{cases}$$

Exemplo

$$\begin{cases} 3x_1 - x_2 - x_3 = 1 \\ x_1 + 3x_2 + x_3 = 5 \\ x_1 - x_2 + 2x_3 = 2 \end{cases}$$

Exemplo

$$\begin{cases} 3x_1 - x_2 - x_3 = 1 \\ x_1 + 3x_2 + x_3 = 5 \\ x_1 - x_2 + 2x_3 = 2 \end{cases}$$



$$\begin{cases} x_1^{(k+1)} = (1 + x_2^{(k)} + x_3^{(k)}) / 3 \\ x_2^{(k+1)} = (5 - x_1^{(k)} - x_3^{(k)}) / 3 \\ x_3^{(k+1)} = (2 - x_1^{(k)} + x_2^{(k)}) / 2 \end{cases}$$

Exemplo

$$\begin{cases} 3x_1 - x_2 - x_3 = 1 \\ x_1 + 3x_2 + x_3 = 5 \\ x_1 - x_2 + 2x_3 = 2 \end{cases}$$



$$\begin{cases} x_1^{(k+1)} = (1 + x_2^{(k)} + x_3^{(k)}) / 3 \\ x_2^{(k+1)} = (5 - x_1^{(k)} - x_3^{(k)}) / 3 \\ x_3^{(k+1)} = (2 - x_1^{(k)} + x_2^{(k)}) / 2 \end{cases}$$

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$	$ x_1^{(k+1)} - x_1^{(k)} $	$ x_2^{(k+1)} - x_2^{(k)} $	$ x_3^{(k+1)} - x_3^{(k)} $
0	0	0	0	-	-	-
1	0.333	1.667	1	0.333	1.667	1
2	1.222	1.222	1.667	0.889	0.555	0.667
3	1.296	0.704	1	0.074	0.518	0.667
4	0.901	0.901	0.704	0.395	0.197	0.296
5	0.868	1.132	1	0.033	0.197	0.296
6	1.044	1.044	1.132	0.176	0.088	0.132

Exemplo

$$\begin{cases} 3x_1 - x_2 - x_3 = 1 \\ x_1 + 3x_2 + x_3 = 5 \\ x_1 - x_2 + 2x_3 = 2 \end{cases}$$



$$\begin{cases} x_1^{(k+1)} = (1 + x_2^{(k)} + x_3^{(k)}) / 3 \\ x_2^{(k+1)} = (5 - x_1^{(k)} - x_3^{(k)}) / 3 \\ x_3^{(k+1)} = (2 - x_1^{(k)} + x_2^{(k)}) / 2 \end{cases}$$

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$	$ x_1^{(k+1)} - x_1^{(k)} $	$ x_2^{(k+1)} - x_2^{(k)} $	$ x_3^{(k+1)} - x_3^{(k)} $
0	0	0	0	-	-	-
1	0.333	1.667	1	0.333	1.667	1
2	1.222	1.222	1.667	0.889	0.555	0.667
3	1.296	0.704	1	0.074	0.518	0.667
4	0.901	0.901	0.704	0.395	0.197	0.296
5	0.868	1.132	1	0.033	0.197	0.296
6	1.044	1.044	1.132	0.176	0.088	0.132

O Método de Gauss-Seidel

Diagram illustrating the Gauss-Seidel method. The initial values $x_1^{(0)}, \dots, x_n^{(0)}$ are used to update the variables $x_1, x_2, \dots, x_n$ iteratively. The equations shown are:

$$\begin{aligned}x_1 &= \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\x_2 &= \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\\vdots &\vdots \\x_n &= \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}]\end{aligned}$$

O Método de Gauss-Seidel

$x_1^{(1)}$ $\left\{ \begin{array}{l} x_1 = \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\ x_2 = \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\ \vdots \\ x_n = \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}] \end{array} \right.$

$x_1^{(0)}, \dots, x_n^{(0)}$

O Método de Gauss-Seidel

Diagram illustrating the Gauss-Seidel method iteration. The initial guess vector $x_1^{(0)}, \dots, x_n^{(0)}$ is used to calculate the next iteration values. The equations are:

$$\begin{aligned} x_1 &= \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\ x_2 &= \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\ &\vdots \\ x_n &= \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}] \end{aligned}$$

The diagram shows the flow of information: the initial guess $x_1^{(0)}, \dots, x_n^{(0)}$ is used to calculate the next iteration values. The updated value $x_1^{(1)}$ is used in the second equation. The terms $a_{21}x_1$, $a_{23}x_3$, and $a_{2n}x_n$ are circled in red, indicating the current iteration values used in the calculation.

O Método de Gauss-Seidel

The diagram illustrates the Gauss-Seidel method for solving a system of linear equations. It shows the iterative update of variables x_1 and x_2 from their initial values $x_1^{(0)}, \dots, x_n^{(0)}$.

The system of equations is presented in a box:

$$\begin{aligned} x_1 &= \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\ x_2 &= \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\ &\vdots \\ x_n &= \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}] \end{aligned}$$

Arrows indicate the flow of information:

- A box at the top contains the initial values $x_1^{(0)}, \dots, x_n^{(0)}$. Dashed arrows point from these values to the terms x_1 , x_3 , and x_n in the equation for x_2 .
- A box on the left contains the updated value $x_1^{(1)}$. A dashed arrow points from this box to the term x_1 in the equation for x_2 .
- A bracket on the left groups the equations for x_2 and x_3 , indicating that x_2 is updated before x_3 .
- Red circles highlight the terms x_1 , x_3 , and x_n in the equation for x_2 , showing that these values are updated as soon as they are available.

O Método de Gauss-Seidel

$$x_1^{(0)}, \dots, x_n^{(0)}$$

The diagram illustrates the Gauss-Seidel method for solving a system of linear equations. It shows a set of equations for variables $x_1, x_2, \dots, x_n$. The equations are:

$$\begin{aligned} x_1 &= \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\ x_2 &= \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\ &\vdots \\ x_n &= \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}] \end{aligned}

On the left, the variables $x_1^{(1)}$, $x_2^{(1)}$, and $x_n^{(1)}$ are shown in boxes. Dashed lines connect these boxes to the corresponding variables in the equations. For example, a dashed line from $x_1^{(1)}$ points to x_1 in the first equation, and another from $x_2^{(1)}$ points to x_2 in the second equation. A bracket groups the equations for $x_1, x_2, \dots, x_n$, with a dashed line from $x_n^{(1)}$ pointing to the group. In the equation for x_n , the terms $a_{n1}x_1$, $a_{n2}x_2$, and $a_{n,n-1}x_{n-1}$ are circled in red, indicating that the values of $x_1, x_2, \dots, x_{n-1}$ are updated and used in the calculation of x_n .$$

O Método de Gauss-Seidel

$x_1^{(1)}$
 $x_2^{(1)}$
 $\vdots$
 $x_n^{(1)}$

$x_1 = \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n]$
 $x_2 = \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n]$
 $\vdots$
 $x_n = \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}]$

$x_1^{(1)}, \dots, x_n^{(1)}$

O Método de Gauss-Seidel

$x_1^{(1)}, \dots, x_n^{(1)}$

$$\begin{aligned} x_1 &= \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\ x_2 &= \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\ &\vdots \\ x_n &= \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}] \end{aligned}$$

O Método de Gauss-Seidel

Diagram illustrating the Gauss-Seidel method. A box at the top contains the previous iteration's values $x_1^{(1)}, \dots, x_n^{(1)}$. Dashed arrows point from these values to the corresponding terms in the equations of the current iteration. The first equation is highlighted with a bracket and a box around $x_1^{(2)}$.

$$\begin{aligned} x_1^{(2)} &= \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\ x_2 &= \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\ &\vdots \\ x_n &= \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}] \end{aligned}$$

O Método de Gauss-Seidel

Diagram illustrating the Gauss-Seidel method iteration. The equations are:

$$\begin{aligned}x_1 &= \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\x_2 &= \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\\vdots &\vdots \\x_n &= \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}]\end{aligned}$$

The diagram shows the flow of data between the equations and the iteration steps. The box at the top contains the previous iteration's values $x_1^{(1)}, \dots, x_n^{(1)}$. The box on the left contains the current iteration's value $x_1^{(2)}$. The equations are shown with the terms x_1 , x_3 , and x_n circled in red, indicating they are updated using the latest available values.

O Método de Gauss-Seidel

Diagram illustrating the Gauss-Seidel method. The system of equations is shown within a box:

$$\begin{aligned}x_1 &= \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\x_2 &= \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\&\vdots \\x_n &= \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}]\end{aligned}$$

Variables are labeled in boxes:

- $x_1^{(2)}$ and $x_2^{(2)}$ are shown on the left, with a bracket indicating they are part of the current iteration's update.
- $x_1^{(1)}, \dots, x_n^{(1)}$ are shown at the top, representing the previous iteration's values.

Dashed arrows indicate the flow of values from the previous iteration to the current iteration. Red circles highlight the terms $a_{21}x_1$, $a_{23}x_3$, and $a_{2n}x_n$ in the second equation, which are updated using the latest values of x_1 , x_3 , and x_n from the previous iteration.

O Método de Gauss-Seidel

$$x_1^{(1)}, \dots, x_n^{(1)}$$

The diagram illustrates the Gauss-Seidel iteration for the n th equation. On the left, a column of variables $x_1^{(2)}$, $x_2^{(2)}$, and $x_n^{(2)}$ is shown, each enclosed in a box. Dashed lines connect these boxes to the corresponding terms in the equation for x_n on the right. Specifically, dashed lines connect $x_1^{(2)}$ to $a_{n1}x_1$, $x_2^{(2)}$ to $a_{n2}x_2$, and $x_n^{(2)}$ to $a_{nn}x_n$. The equation for x_n is
$$x_n = \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}]$$
 The terms x_1 , x_2 , and x_{n-1} in the brackets are circled in red. Arrows point from the circled x_1 and x_2 to the circled x_{n-1} , indicating that the most recent values of x_1 and x_2 are used in the calculation of x_n . The variable x_n on the left side of the equation is also circled in red.

$$\begin{array}{l} x_1^{(2)} \\ x_2^{(2)} \\ \vdots \\ x_n^{(2)} \end{array} \quad \begin{array}{l} x_1 = \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n] \\ x_2 = \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n] \\ \vdots \\ x_n = \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}] \end{array}$$

O Método de Gauss-Seidel

$x_1^{(2)}$
 $x_2^{(2)}$
 $\vdots$
 $x_n^{(2)}$

$x_1 = \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n]$
 $x_2 = \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n]$
 $\vdots$
 $x_n = \frac{1}{a_{nn}} [b_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}]$

$x_1^{(2)}, \dots, x_n^{(2)}$

Comparação

Gauss-Jacobi

$$\begin{aligned}x_1^{(1)} &= \frac{1}{a_{11}} [b_1 - a_{12}x_2^{(0)} - a_{13}x_3^{(0)} - \dots - a_{1n}x_n^{(0)}] \\x_2^{(1)} &= \frac{1}{a_{22}} [b_2 - a_{21}x_1^{(0)} - a_{23}x_3^{(0)} - \dots - a_{2n}x_n^{(0)}] \\&\vdots \\x_n^{(1)} &= \frac{1}{a_{nn}} [b_n - a_{n1}x_1^{(0)} - a_{n2}x_2^{(0)} - \dots - a_{n,n-1}x_{n-1}^{(0)}]\end{aligned}$$

Gauss-Seidel

$$\begin{aligned}x_1^{(1)} &= \frac{1}{a_{11}} [b_1 - a_{12}x_2^{(0)} - a_{13}x_3^{(0)} - \dots - a_{1n}x_n^{(0)}] \\x_2^{(1)} &= \frac{1}{a_{22}} [b_2 - a_{21}x_1^{(1)} - a_{23}x_3^{(0)} - \dots - a_{2n}x_n^{(0)}] \\&\vdots \\x_n^{(1)} &= \frac{1}{a_{nn}} [b_n - a_{n1}x_1^{(1)} - a_{n2}x_2^{(1)} - \dots - a_{n,n-1}x_{n-1}^{(1)}]\end{aligned}$$

Exemplo

$$\begin{cases} 3x_1 - x_2 - x_3 = 1 \\ x_1 + 3x_2 + x_3 = 5 \\ x_1 - x_2 + 2x_3 = 2 \end{cases}$$



$$\begin{cases} x_1^{(k+1)} = (1 + x_2^{(k)} + x_3^{(k)}) / 3 \\ x_2^{(k+1)} = (5 - x_1^{(k+1)} - x_3^{(k)}) / 3 \\ x_3^{(k+1)} = (2 - x_1^{(k+1)} + x_2^{(k+1)}) / 2 \end{cases}$$

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$	$ x_1^{(k+1)} - x_1^{(k)} $	$ x_2^{(k+1)} - x_2^{(k)} $	$ x_3^{(k+1)} - x_3^{(k)} $
0	0	0	0	-	-	-
1	0.333	1.555	1.611	0.333	1.555	1.611
2	1.388	0.666	0.638	1.055	0.888	0.972
3	0.768	1.197	1.214	0.620	0.531	0.575
4	1.137	0.882	0.872	0.368	0.314	0.341
5	0.918	1.069	1.075	0.218	0.186	0.202
6	1.048	0.958	0.955	0.129	0.110	0.120

Exemplo

$$\begin{cases} 3x_1 - x_2 - x_3 = 1 \\ x_1 + 3x_2 + x_3 = 5 \\ x_1 - x_2 + 2x_3 = 2 \end{cases}$$



$$\begin{cases} x_1^{(k+1)} = (1 + x_2^{(k)} + x_3^{(k)}) / 3 \\ x_2^{(k+1)} = (5 - x_1^{(k+1)} - x_3^{(k)}) / 3 \\ x_3^{(k+1)} = (2 - x_1^{(k+1)} + x_2^{(k+1)}) / 2 \end{cases}$$

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$	$ x_1^{(k+1)} - x_1^{(k)} $	$ x_2^{(k+1)} - x_2^{(k)} $	$ x_3^{(k+1)} - x_3^{(k)} $
0	0	0	0	-	-	-
1	0.333	1.555	1.611	0.333	1.555	1.611
2	1.388	0.666	0.638	1.055	0.888	0.972
3	0.768	1.197	1.214	0.620	0.531	0.575
4	1.137	0.882	0.872	0.368	0.314	0.341
5	0.918	1.069	1.075	0.218	0.186	0.202
6	1.048	0.958	0.955	0.129	0.110	0.120

Critérios de Convergência

Se o sistema $\mathbf{Ax} = \mathbf{b}$ tiver **diagonal dominante**, os métodos Gauss-Jacobi e Gauss-Seidel convergem para a solução do sistema.

$$\alpha_i = \frac{\sum_{j=1, j \neq i}^n |a_{ij}|}{|a_{ii}|}$$

$$\alpha = \max(\alpha_i) < 1$$

Critérios de Convergência

Se o sistema $\mathbf{Ax} = \mathbf{b}$ tiver **diagonal dominante**, os métodos Gauss-Jacobi e Gauss-Seidel convergem para a solução do sistema.

Exemplo:

$$\begin{cases} 3x_1 - x_2 - x_3 = 1 & |3| \geq |-1| + |-1| & V \\ x_1 + 3x_2 + x_3 = 5 & |3| \geq |1| + |1| & V \\ x_1 - x_2 + 2x_3 = 2 & |2| \geq |1| + |-1| & V \end{cases}$$

Critérios de Convergência

Para o método de Gauss-Seidel, o critério pode ser relaxado (**Critério de Sassenfeld**).

$$\beta_i = \frac{\sum_{j=1}^{i-1} |a_{ij}| \beta_j + \sum_{j=i+1}^n |a_{ij}|}{|a_{ii}|}$$

$$\beta = \max(\beta_i) < 1$$

Critérios de Convergência

Para o método de Gauss-Seidel, o critério pode ser relaxado (**Critério de Sassenfeld**).

Exemplo:

$$\left\{ \begin{array}{ll} 3x_1 - x_2 - x_3 = 1 & |3| \geq |-1| + |-1| \quad V \\ x_1 + 3x_2 + x_3 = 5 & |3| \geq |1| + |1| \quad V \\ x_1 - x_2 + 2x_3 = 2 & |2| \geq |1| + |-1| \quad V \end{array} \right. \Rightarrow 2/3 = 0,6666 = \beta_1$$

Critérios de Convergência

Para o método de Gauss-Seidel, o critério pode ser relaxado (**Critério de Sassenfeld**).

Exemplo:

$$\left\{ \begin{array}{ll} 3x_1 - x_2 - x_3 = 1 & |3| \geq |-1| + |-1| \quad V \rightarrow 2/3 = 0,6666 = \beta_1 \\ x_1 + 3x_2 + x_3 = 5 & |3| \geq |1| + |1| \quad V \rightarrow (0,6666+1) \\ x_1 - x_2 + 2x_3 = 2 & |2| \geq |1| + |-1| \quad V \end{array} \right.$$

Critérios de Convergência

Para o método de Gauss-Seidel, o critério pode ser relaxado (**Critério de Sassenfeld**).

Exemplo:

$$\left\{ \begin{array}{ll} 3x_1 - x_2 - x_3 = 1 & |3| \geq |-1| + |-1| \quad V \\ x_1 + 3x_2 + x_3 = 5 & |3| \geq |1| + |1| \quad V \\ x_1 - x_2 + 2x_3 = 2 & |2| \geq |1| + |-1| \quad V \end{array} \right. \begin{array}{l} \rightarrow 2/3 = 0,6666 = \beta_1 \\ \rightarrow (0,6666 + 1)/3 \\ = 0,5555 = \beta_2 \end{array}$$

Critérios de Convergência

Para o método de Gauss-Seidel, o critério pode ser relaxado (**Critério de Sassenfeld**).

Exemplo:

$$\begin{cases} 3x_1 - x_2 - x_3 = 1 & |3| \geq |-1| + |-1| & V \rightarrow 2/3 = 0,6666 = \beta_1 \\ x_1 + 3x_2 + x_3 = 5 & |3| \geq |1| + |1| & V \rightarrow (0,6666 + 1)/3 \\ & & = 0,5555 = \beta_2 \\ x_1 - x_2 + 2x_3 = 2 & |2| \geq |1| + |-1| & V \rightarrow (0,6666 + 0,5555)/2 \\ & & = 0,61105 = \beta_3 \end{cases}$$

Critérios de Convergência

Se os critérios de convergência forem satisfeitos, os métodos Gauss-Jacobi e Gauss-Seidel convergem para a solução do sistema para qualquer $\mathbf{X}^{(0)}$.

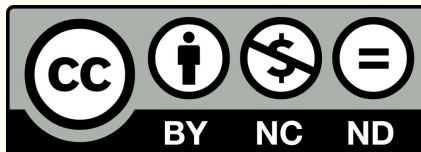
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